Active Maintenance Report: coppock_green_porter_2022

Coppock, Green and Porter (2022, Research & Politics): Does Digital Advertising Affect Vote Choice?

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This repository holds the actively maintained replication code for Coppock, Green and Porter (2022), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1177/20531680221076901
Replication archive	10.7910/DVN/UH2NQW
Pre-analysis plan	egap.org/registration/5312, mirrored at osf.io/ch4ms

The data are not redistributed here. The deposit is 1.9 MB across 14 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

To reproduce. Clone or download the repository, open `coppock_green_porter_2022.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 14 files, and produces every table and figure into `maintained/output/`. Required packages: tidyverse, estimatr, modelsummary, DeclareDesign, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about two minutes, nearly all of it in the two simulation scripts: 2000 re-randomizations for the randomization inference and 5000 design simulations for Figure G.2. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Almost. Eight of the nine analysis scripts execute without error on a current R installation. The ninth, `CGP_2022_equivalence_tests.R`, stops on its first model: it asks `lh_robust()` for a linear combination of coefficients with clustered CR2 standard errors, and current `estimatr` refuses, because `lh_robust()` has no CR2 path. Everything in appendix Table C.6 therefore has no live source in the deposit, which is 30 of the 210 recorded quantities.

Of what does run, every number matches the table it belongs to. Two numbers in the article's prose do not match the table they describe, but the code is right and the sentence is wrong; see the errata. The one class of quantity that cannot reproduce exactly is the simulated one: neither the randomization inference nor the design diagnosis sets a seed, so the archive does not return its own published p-values twice in a row, let alone four years later. Both scripts also ship with the number of simulations cut by a factor of four to ten below the value used for the paper, with the published setting left commented out on the line above.

1.2 Does the maintained rewrite reproduce the paper?

Yes, with the exceptions that are the point of the exercise. 185 of the 192 verifiable ground truth claims match the published values to reported precision. The 7 that do not are these:

- **Two are errors in the published text.** Describing Table 4's first column, the article says the unadjusted estimate is "a 2.1 percentage point increase" with "a standard error of 3.0 percentage points". Table 4 gives 0.0021 with a standard error of 0.0225, which is 0.21 and 2.25 percentage points. The estimate in the text is off by a factor of ten and the standard error does not correspond to any model in the paper. The same sentence appears in the preprint, so it is not a typesetting accident, and it inflates the reported effect tenfold in the only place a reader meets it in prose.
- **Four are simulation draws.** The two randomization inference p-values and the two power figures differ from the published values by less than one Monte Carlo standard error. They cannot be matched exactly, because the archive sets no seed.
- **One is a figure label.** The published Figure 1 annotates this study's standard error as 0.009 because the archive typed the table's rounded 0.0085 into the figure. The rewrite reads the estimate from the Table 4 output instead, and 0.0084760 rounds to 0.008. Every posterior in the figure is unchanged.

The remaining 18 recorded quantities are unverifiable rather than unmatched: 12 belong to a table the appendix never printed (see the errata), four are precinct counts for a sample the deposit does not contain, and two are advertisement-vendor exposure counts that were never deposited in any form.

2 Paper overview

Citation: Coppock, A., Green, D. P. and Porter, E. (2022). "Does digital advertising affect vote choice? Evidence from a randomized field experiment." *Research & Politics*, 9(1). DOI: 10.1177/20531680221076901

Summary: A blocked, clustered field experiment run during the 2018 midterms tests whether pre-roll video advertisements on Facebook and Instagram move vote choice. Two hundred and ten Florida ZIP codes across four congressional districts were grouped into matched trios; one ZIP in each trio was assigned to treatment and then to one of two anti-Republican advertisements on gun policy, neither of which named a candidate. The advertisements ran from 25 October 2018 until each exhausted a $30,000 budget, generating 1.1 million three-second views. The outcome is the precinct-level Democratic two-party vote share from the Florida Secretary of State, analysed on the 853 precincts that fall wholly inside a study ZIP code and have outcome data, with CR2 standard errors clustered by ZIP code and one-tailed randomization inference under the sharp null. The unadjusted estimate is 0.21 percentage points (SE 2.25); adjusting for the 2012, 2014 and 2016 vote shares brings it to -0.04 points (SE 0.85). A Bayesian integration of this result with the three prior field experiments on digital advertising leaves a posterior centred on 0.7 percentage points with a standard deviation of 0.4.

3 Original archive reproducibility

Archive: Harvard Dataverse, 10.7910/DVN/UH2NQW, published 5 January 2022. Fourteen files inside a `replication_archive` directory: nine analysis scripts, two `.rds` data files, a README, a session log, and an `.Rapp.history` file carrying one line of a coauthor's local path. The README names `tidyverse`, `DeclareDesign`, `xtable` and `texreg` as the required add-ons and records R 4.1.1 under macOS Big Sur.

Table 1: Original archive, run script by script in a

Script	Status
CGP_2022_in_text_calculations.R	Clean
CGP_2022_precinct_level_analysis.R	Clean
CGP_2022_zip_code_level_analysis.R	Clean
CGP_2022_precinct_level_analysis_CD.R	Clean
CGP_2022_design_tests.R	Clean
CGP_2022_bayesian_learning.R	Clean
CGP_2022_precinct_level_randomization_inference.R	Runs, but unseeded and at a quarter of the publish
CGP_2022_design_diagnosis.R	Runs, but unseeded and at a tenth of the published

CGP_2022_equivalence_tests.R	Fails on the first model

Classification: resolvable. One script fails and the fix is arithmetic rather than a change of method.

The failure. `lh_robust()` fits the model and then evaluates a linear hypothesis on the fitted coefficients. Current `estimatr` will not do this with CR2 standard errors, which are what `clusters =` implies, so all six of the script's models stop with `lh_robust not available for CR2 standard errors`. The quantity wanted is a difference between two coefficients in a model that `lm_robust()` fits without complaint, so the rewrite takes the difference and its standard error from the fitted coefficient vector and CR2 variance-covariance matrix directly. That is the same arithmetic `lh_robust()` performed, and it returns Table C.6 exactly.

Two claims in the archive that turn out to be false, or at least stale. Both simulation scripts carry a line of the form `# simulations <- simulate_design(design, sims = 2000)` immediately above the line that runs with `sims = 500`. The implication is that the published numbers came from the larger run and that the smaller one is there for the replicator's convenience. Measured, the larger runs cost 32 seconds and 55 seconds on a 2026 laptop, so there is no convenience to buy. The rewrite runs at the published simulation counts.

Style-level issues, none fatal. `rm(list = ls())` opens all nine scripts. `%>%` throughout. `do(tibble(...))` in the Bayesian learning script, superseded by `reframe()`. `texreg()` and `print.xtable()` write LaTeX to standard output, so the tables in the paper were produced by copying console output rather than by writing a file. Every path in the scripts is bare, with a `# setwd() to replication archive` comment standing in for the working directory.

One deposited file is not part of the analysis. `.Rapp.history` contains a single `load()` call pointing at `/Users/eporter/Dropbox/...` and a file, `CGP_2022_precinct_level_ep_rep.rds`, that is not in the deposit. It is a stray console history, harmless but worth noticing, since it is the only trace of the working directory the analysis was actually run in.

4 Errata

Four defects in the published record, none of which changes a substantive conclusion, all of which the rewrite corrects or flags.

1. The text misstates the unadjusted estimate by a factor of ten. "The first column shows the unadjusted difference-in-means estimate of the treatment effect of any video: a 2.1 percentage point increase in Democratic vote share, though this estimate is uncertain, as evidenced by its large standard error of 3.0 percentage points." Table 4 reports 0.0021 (0.0225). Converted to percentage points, that is 0.21 with a standard error of 2.25. The estimate is stated at ten times its value, and the standard error matches no model in the paper or appendix. The sentence's conclusion is unaffected, since both readings describe an estimate that is small and swamped by its standard error, but the number itself is wrong in the article and in the preprint.

2. Appendix Table B.3 prints Table B.4. The caption of B.3 reads "Effects on vote share (CD fixed effects)", and every cell beneath it is identical to Table B.4, "Effects on vote margin (CD fixed effects)": the same coefficients, standard errors, R-squared values and covariate labels, all on the vote margin scale. The vote share models with congressional district fixed effects are in the archive but appear nowhere in the appendix. The rewrite computes them; `maintained/output/table_b3_vote_share_precinct_cd.csv` is where they live. The estimate on any treatment video is 0.0042 (0.0217) unadjusted and -0.0018 (0.0088) adjusted, which is the same substantive story as Table 4.

3. Covariate row labels are wrong in five appendix tables. Tables E.7 and F.8 regress the treatment indicator on lagged two-party *vote margins* and label the rows "Two Party Vote Share". Tables F.9, F.10 and F.11 pass `texreg` a coefficient-name vector one label short of the model, so every covariate row from 2016 down is labelled with the name belonging to the row above it: the 2016 vote share coefficient is printed as "Missingness Indicator (2016)". No coefficient moves; only its name does. The rewrite labels each row with the variable actually in the model.

4. Figure G.2's panel annotation names the wrong effect size. The label reads "Power when PATE = 0.1". The simulated effects are drawn from a normal centred at 0.01, which is also what the appendix text says two paragraphs above. The string is hardcoded in the archive script. The rewrite prints 0.01.

5 Ground truth

Table 2: Ground truth, by published object

Table or figure	Claims	Archive verifiable	Archive matches	Rewrite verifiable	Rewrite matches
figure_1	12	12	12	12	11
figure_g2	2	2	0	2	0
table_1	24	20	20	20	20
table_4	14	14	14	14	14
table_a1	14	14	14	14	14
table_a2	13	13	13	13	13
table_b3	12	0	0	0	0
table_b4	13	13	13	13	13
table_b5	13	13	13	13	13
table_c6	30	0	0	30	30
table_e7	6	6	6	6	6

table_f10	13	13	13	13	13
table_f11	13	13	13	13	13
table_f8	5	5	5	5	5
table_f9	13	13	13	13	13
text	13	11	7	11	7

Every value in the `value_paper` column of `ground_truth/coppock_green_porter_2022_ground_truth.csv` was read off the published article or its appendix. Where the paper does not state a quantity, the column is blank and the row is marked unverifiable rather than matched.

Table 3: The 7 claims the rewrite does not match

Claim	Published	Rewrite
figure_1: present_study_se_label	0.009	0.0085
figure_g2: power_ols	0.210	0.2072
figure_g2: power_posterior	0.893	0.8866
text: unadjusted_estimate_pp	2.100	0.2114
text: unadjusted_se_pp	3.000	2.2487
text: ri_p_unadjusted	0.471	0.4585
text: ri_p_adjusted	0.508	0.5050

6 Maintained rewrite

Table 4: Maintained rewrite: one script per published object

Script	Produces
helpers.R	Packages, the cluster-count glance method, the table footer spec
table_1_sample_description.R	Table 1
table_4_vote_share_precinct.R	Table 4
table_a1_vote_margin_precinct.R	Table A.1
table_a2_turnout_precinct.R	Table A.2
table_b3_vote_share_precinct_cd.R	Table B.3, as it should have appeared
table_b4_vote_margin_precinct_cd.R	Table B.4
table_b5_turnout_precinct_cd.R	Table B.5
table_c6_equivalence_tests.R	Table C.6
table_e7_balance_precinct.R	Table E.7 and its joint F test
table_f8_balance_zip.R	Table F.8 and its joint F test
table_f9_vote_share_zip.R	Table F.9

table_f10_vote_margin_zip.R	Table F.10
table_f11_turnout_zip.R	Table F.11
figure_1_bayesian_learning.R	Figure 1
figure_g2_design_diagnosis.R	Figure G.2
text_in_text_calculations.R	The vote totals, posteriors and cost per vote quoted in the text
text_randomization_inference.R	The two randomization inference p-values quoted in the text

Every script writes a full-precision `.csv` of every coefficient it estimates and, for the regression tables, a `.tex` rendering with the same rows and columns as the published table, so a reader can compare them line by line.

Substitutions.

Archive	Rewrite	Why
`lh_robust()` with clusters	`lm_robust()` plus the contrast taken from the CR2 variance-covariance matrix	`lh_robust()` has no CR2 path; the arithmetic is identical
`texreg()` and `print.xtable()` to standard output	`modelsummary(output = )`	The table is written to a file rather than copied out of a console
`do(tibble(...))`	`reframe()`	`do()` is superseded
`DeclareDesign:::format_num()`	`sprintf()`	Unexported internals are not a stable interface
`separate()`	`separate_wider_delim()`	`separate()` is superseded
`%>%`, `ifelse()`, `rm(list = ls())`	`|>`, `if_else()`, omitted	House style
`sims = 500`, no seed	`sims = 2000` and `sims = 5000`, seeded	The published simulation counts, at a measured cost of 32 and 55 seconds

No published number is an input to a computation. The archive's Figure 1 script typed this paper's own estimate and standard error into the vector of experiments being pooled, and the in-text script typed the vote total and the posterior mean into the cost-per-vote formula. In the rewrite, `figure_1_bayesian_learning.R` reads the estimate out of `output/table_4_vote_share_precinct.csv`, and `text_in_text_calculations.R` reads the posterior out of `output/figure_1_bayesian_learning.csv` and the vote total out of the ZIP-level data. The only constants entered by hand are the three prior studies' published estimates, which are other people's results and belong in the code, and the campaign's budget, which is a design fact rather than an estimate.

The cost per vote depends on which posterior you use. The article's footnote computes $100{,}000/(822{,}783 \times 0.007 \times 2) = \$8.68$ from the rounded quantities it quotes. The unrounded posterior is 0.00708, and the same calculation at full precision gives $8.58. The rewrite reports both, and the ground truth compares the first, since that is the number in the paper.

7 Figure verification

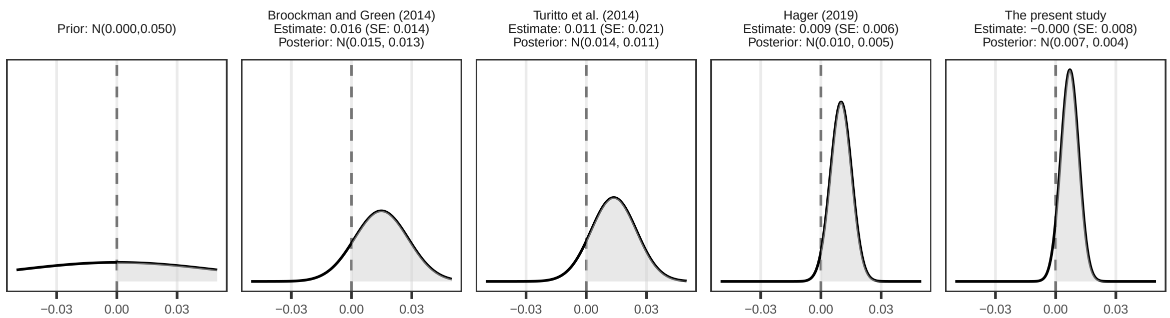

Figure 1: Figure 1, maintained rewrite. Compare against the published figure at doi 10.1177/20531680221076901.

Table 5: Figure 1, panel by panel

Panel	Estimate entering the update	Posterior mean	Posterior SD
Prior		0.0000	0.0500
Broockman and Green (2014)	0.0160	0.0148	0.0135
Turitto et al. (2014)	0.0110	0.0137	0.0113
Hager (2019)	0.0090	0.0100	0.0053
The present study	-0.0004	0.0071	0.0045

Every posterior matches the published panel labels: N(0.015, 0.013), N(0.014, 0.011), N(0.010, 0.005) and N(0.007, 0.004). The one visible difference is the standard error annotated on the last panel, which is 0.008 here and 0.009 in the article, for the reason given in the summary.

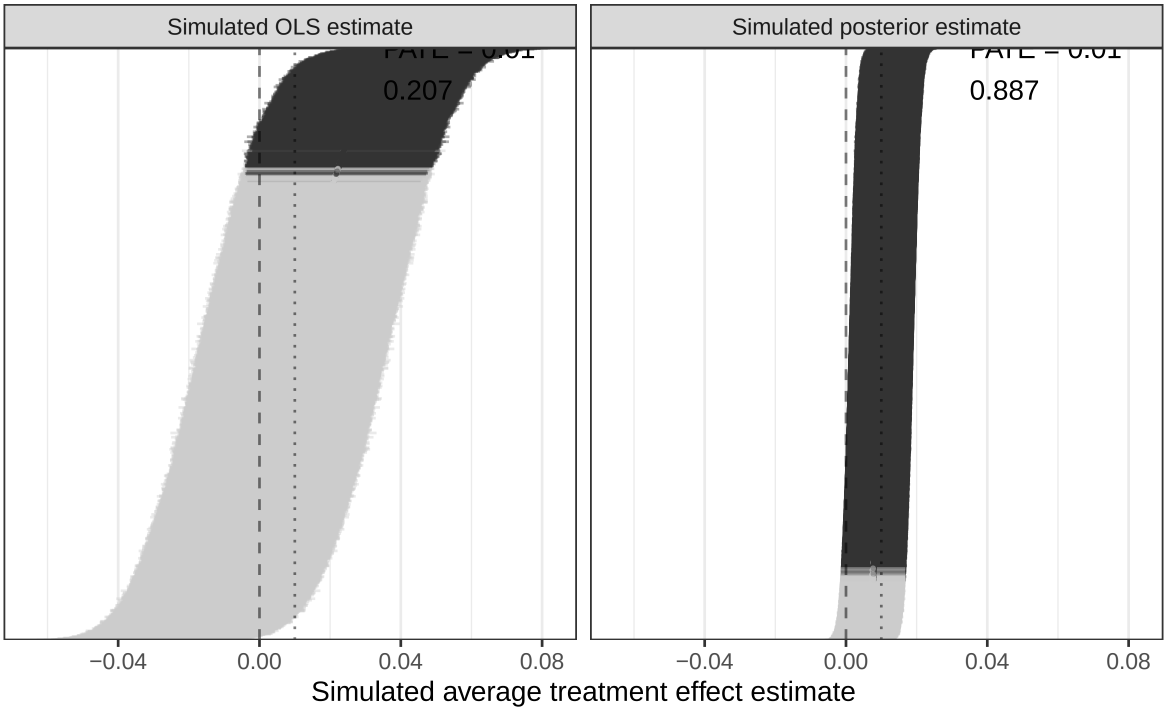

Figure 2: Figure G.2, maintained rewrite, at the published 5000 simulations.

Table 6: Figure G.2 diagnosands

Simulations	Mean OLS estimate	Power, OLS	Power, posterior
5000	0.0097	0.207	0.887

The appendix reports power of 0.210 for the OLS estimator and 0.893 for the posterior. Both are within one Monte Carlo standard error of the values above, which is as close as an unseeded simulation permits.

8 Rewrite verification

Table 7: Rewrite against the published values

Table or figure	Claims	Matching	Not matching
figure_1	12	11	1
figure_g2	2	0	2

table_1	20	20	0
table_4	14	14	0
table_a1	14	14	0
table_a2	13	13	0
table_b4	13	13	0
table_b5	13	13	0
table_c6	30	30	0
table_e7	6	6	0
table_f10	13	13	0
table_f11	13	13	0
table_f8	5	5	0
table_f9	13	13	0
text	11	7	4

Beyond the ground truth, the `.tex` files in `maintained/output/` reproduce the published tables cell for cell, including the covariate rows, the R-squared values, the sample sizes and the significance stars. Table 4, Tables A.1 and A.2, Tables B.4 and B.5, Tables E.7 and F.8, and Tables F.9 through F.11 all match the published tables in every printed cell.

Determinism. Every script other than the two simulation scripts is deterministic and returns byte-identical output on a second run. The two simulation scripts are seeded, so they are also reproducible run to run; they are not reproducible against the archive, which set no seed.

9 R environment

Table 8: Versions this report was built against

Component	Version
R	4.6.0
tidyverse	2.0.0
estimatr	1.0.6
modelsummary	2.6.0
DeclareDesign	1.1.1
knitr	1.51
kableExtra	1.4.0
here	1.0.2

The archive was written against R 4.1.1 with `estimatr` 0.30.2 and `DeclareDesign` 0.28.0. The `lh_robust()` failure is the only place where the four intervening years of package development broke something outright.